



Volcanic vapor releases arsenic into the atmosphere.

Explosive gases

Many minerals in Earth's crust have high concentrations of arsenic compounds, and volcanic gases push these out into the atmosphere. When a volcano erupts, it releases a complicated mixture of chemicals at the surface, including a lot of arsenic. This can contaminate the air or nearby groundwater sources.

Arsenic

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As

One of the few elements that exhibit the properties of both metals and non-metals, arsenic is a metalloid. It is found in many minerals in Earth's crust, often combined with oxygen, chlorine, or sulfur. Arsenic rightly has a reputation as a deadly poison. However, some arsenic compounds have genuine medicinal uses, for example arsenic trioxide is used to treat a certain kind of blood cancer.

Cluster of realgar crystals



Realgar

This arsenic sulfide mineral is found in hot volcanic springs and is an important source of arsenic. Realgar is toxic and has been used as a rat poison and a weedkiller.

DETECTING ARSENIC

In the early 1800s, arsenic compounds, which are tasteless and odorless, were sometimes used to poison food or drink without being detected. In 1836, the English chemist James Marsh devised a method (shown here) of detecting even the tiniest amounts of arsenic in food. This was called the Marsh Test.

